

## Total bodyweight and sex both drive pharmacokinetic variability of fluconazole in obese adults

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**Background:** Fluconazole is commonly used to treat or prevent fungal infections. It is typically used orally but in critical situations, IV administration is needed. Obesity may influence the pharmacokinetics and therapeutic efficacy of a drug. In this study, we aim to assess the impact of obesity on fluconazole pharmacokinetics given orally or IV to guide dose adjustments for the obese population.

**Methods:** We performed a prospective pharmacokinetic study with intensive sampling in obese subjects undergoing bariatric surgery ( $n=17$ , BMI  $\geq 35$  kg/m<sup>2</sup>) and non-obese healthy controls ( $n=8$ ,  $18.5 \leq \text{BMI} < 30.0$  kg/m<sup>2</sup>). Participants received a semi-simultaneous oral dose of 400 mg fluconazole capsules, followed after 2 h by 400 mg IV. Population pharmacokinetic modelling and simulation were performed using NONMEM 7.3.

**Results:** A total of 421 fluconazole concentrations in 25 participants (total bodyweight 61.0–174 kg) until 48 h after dosing were obtained. An estimated bioavailability of 87.5% was found for both obese and non-obese subjects, with a 95% distribution interval of 43.9%–98.4%. With increasing total bodyweight, both higher CL and  $V_d$  were found. Sex also significantly impacted  $V_d$ , being 27% larger in male compared with female participants.

**Conclusions:** In our population of obese but otherwise healthy individuals, obesity clearly alters the pharmacokinetics of fluconazole, which puts severely obese adults, particularly if male, at risk of suboptimal exposure, for which adjusted doses are proposed.

## Introduction

The prevalence of obesity (BMI  $\geq 30$  kg/m<sup>2</sup>) has nearly tripled over the past 50 years.<sup>1,2</sup> Obese individuals often have an increased risk to develop infections, including fungal infections.<sup>3–5</sup> Obesity is known to influence the pharmacokinetics for many drugs and is associated with underdosing of antimicrobials, which may negatively impact clinical outcomes.<sup>5–7</sup>

Fluconazole is a widely used antifungal agent to prevent and treat *Candida* infections, including invasive candidiasis, and superficial infections such as oropharyngeal candidiasis,

oesophageal candidiasis, candiduria and vaginal candidiasis. Low fluconazole exposure is associated with increased mortality.<sup>8,9</sup> A threshold of  $fAUC_{24\text{ h}}/\text{MIC} > 100$  is recommended to treat invasive candidiasis.<sup>10–12</sup> Due to low plasma protein binding (11%–12%), this can be translated to  $AUC_{24\text{ h}} > 200$  mg·h/L for the *Candida* clinical breakpoint against fluconazole with MIC equal to 2 mg/L.<sup>10</sup>

Fluconazole is available for IV use and as capsules, suspension or tablets for oral administration. In non-obese subjects, the oral bioavailability (F) was reported to be over 90%, but this has not been studied in the obese population.<sup>10</sup> As it is reported that

obesity is associated with increased gut permeability and accelerated gastric emptying, it is possible that obesity could influence the oral absorption of fluconazole.<sup>7</sup> Moreover, with fluconazole being primarily cleared renally, its CL could be affected by obesity, which is associated with increased renal flow.<sup>7</sup>

A dedicated study on the impact of obesity on the pharmacokinetics of fluconazole, in the absence of other potentially confounding patient characteristics, is lacking. This study characterizes the pharmacokinetics, including the oral F and absorption rate of the capsule formulation, in healthy non-obese and otherwise healthy morbidly obese adults. The results are used to derive model-based dosing recommendations for this special population.

## Methods

### Study population

Obese adults with BMI > 35 kg/m<sup>2</sup> undergoing laparoscopic gastric bypass surgery or sleeve gastrectomy at the St. Antonius Hospital (Nieuwegein, The Netherlands) and non-obese healthy volunteers (BMI = 18.5–30.0 kg/m<sup>2</sup>) from the Radboud University Medical Center (Nijmegen, The Netherlands), were included. Participants were eligible for inclusion if they were aged 18–65 years. Participants were excluded if they were allergic to fluconazole or other azoles, pregnant or breast-feeding, taking medication with a known interaction with fluconazole, diagnosed with renal or hepatic dysfunctions, or had a history of long QT syndrome, or drug or alcohol abuse. Written informed consent was obtained before inclusion. This study was approved by the Dutch Medical Research Ethics Committees United (NL66611.100.18) and was conducted in accordance with the Declaration of Helsinki and good clinical practice guidelines (ClinicalTrials.gov identifier: NCT04122560).

### Study design

Participants received a semi-simultaneous oral dose of 400 mg fluconazole as capsules, followed 2 h later by 400 mg IV infusion over approximately 20 min. Eight blood samples were collected after oral administration and nine samples were collected after IV administration up to 48 h after the oral dose, or until discharge for obese participants. Blood samples were collected in heparin tubes, centrifuged at 1900 g for 5 min and stored at –80°C until analysis.

In all individuals, 24 h urine and serum creatinine were collected on the day of study and glomerular filtration rate (GFR) was calculated.<sup>13</sup> Additionally, estimated GFR values were calculated using the four-variable Modification of Diet in Renal Disease (MDRD),<sup>14</sup> the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI)<sup>15</sup> and the conventional Cockcroft–Gault, either calculated with total bodyweight (TBW) or with lean bodyweight (LBW).<sup>16,17</sup> MDRD and CKD-EPI were de-indexed for body surface area (BSA) by multiplying the conventional values (in mL/min/1.73 m<sup>2</sup>) by BSA/1.73.<sup>18</sup>

### Analytical assay

Fluconazole plasma concentrations were measured by a validated assay using LC coupled with tandem MS. Plasma samples were treated with protein precipitation procedures. The lower limit of detection (LOD) was 0.005 mg/L and the lower limit of quantification (LLOQ) and upper limit of quantification were 0.25 and 30.2 mg/L, respectively. The intraday and interday variability were 2.8% and 1.5%, respectively. The assay was externally validated by an international proficiency testing programme.<sup>19,20</sup>

### Population pharmacokinetic model

The population pharmacokinetic model was developed using the non-linear mixed-effects modelling software NONMEM (version 7.3.0; ICON Development Solutions, Hanover, MD, USA) supported by Perl-speaks-NONMEM (version 4.2.0) with the Pirana interface (version 2.9.0; Certara USA, Inc., Princeton, USA).<sup>21</sup> Data pre-processing and visualization were performed with R 4.0.3 and RStudio 1.3.959. The first-order conditional estimation method with interaction was used for all model runs.

For concentrations below the LOD (17 samples, 4.0%), half the LOD (0.0025 mg/L) was imputed. When consecutive samples were below the LOD during the absorption phase, only the last concentration was imputed and the first was omitted. Concentrations between LOD and LLOQ (six samples, 1.4%) were included in the analysis as reported by the lab.

Model development consisted of: (1) selection of the structural model, including disposition and absorption model structures; (2) selection of the statistical error model including inter-individual variability (IIV) and residual unexplained variability (RUV); and (3) covariates analysis. One- and two-compartment disposition models with linear elimination were tested. Tested approaches to describe oral absorption included first-order absorption (with and without absorption lag time), transit compartment models,<sup>22,23</sup> mixed first-order and zero-order absorption<sup>24,25</sup> and a Weibull function.<sup>25</sup> Since peak concentrations were not discernible within 2 h for most individuals, a simulation and re-estimation approach was performed to confirm the identifiability of F, which was then included with a logit function. Proportional, additive and combined additive and proportional error models were assessed for RUV. Covariance between model parameters was assessed and included in the model if correlation coefficients were >0.8.

Model selection was based on the difference in objective function value (OFV, –2 log-likelihood), on the relative standard error of parameter estimates being <50%, physiological plausibility of the parameter estimates, and basic goodness-of-fit (GOF) plots. Particular attention was paid to unbiased description in the oral absorption phase.

Potential covariates were selected based on correlations between empirical Bayes estimates and the covariates in the base model. Tested covariates include sex, age, obesity (as a binary factor), TBW, BMI, BSA, LBW,<sup>26</sup> ideal bodyweight<sup>27</sup> and adjusted bodyweight.<sup>28</sup> The equations for the calculation of the different body size measures can be found in Table S1, available as [Supplementary data](#) at JAC Online. As fluconazole is mainly renally cleared, kidney function-related measures were tested as covariates on CL. Equations for the calculation of these kidney function indices can be found in Table S2. Continuous covariates were tested with linear and power functions centralized for a typical individual of 70 kg for TBW or the median value for the covariate in the dataset. Binary covariates were incorporated with a proportional relationship. Covariate analysis followed a forward inclusion and backward deletion step, with the inclusion criteria of an OFV difference of >3.84 and >6.64, respectively.

The final model was validated using a jackknife analysis and a normalized prediction distribution error (NPDE) analysis based on 1000 simulations. Parameter precision was assessed using the sampling importance resampling method.<sup>29</sup>

### Model-based dosing evaluation and optimization

Stochastic simulations using the final model were performed to illustrate the influence of covariates on fluconazole exposure, to evaluate the currently recommended dosing, and to provide guidance on optimized dosing. Male and female representatives with TBWs of 60, 100, 130 and 170 kg were simulated 1000 times with IIV to predict fluconazole concentration–time profiles and the AUC<sub>24 h</sub>.

For the treatment of invasive fungal infections, a dose of 800 mg on Day 1 followed by a maintenance dose of 400 mg once daily was evaluated.<sup>30</sup> Other loading (Day 1)/maintenance oral dosing regimens that

were evaluated included 400/200 or 200/100 mg, typically used for treating oropharyngeal or oesophageal candidiasis, and 150 mg every third day for a total of three doses (Days 1, 4 and 7) in the first week, and 150 mg weekly for recurrent vaginal candidiasis.<sup>30,31</sup> An  $AUC_{24\text{ h}}$  of  $>200\text{ mg}\cdot\text{h/L}$  was selected as it is a target for the empirical treatment of invasive candidiasis that is not suspected to be located in the brain.<sup>10</sup> An  $AUC_{24\text{ h}}$  of  $>400\text{ mg}\cdot\text{h/L}$  was selected as it is a target for *Candida* meningitis or encephalitis.<sup>32–34</sup>  $AUC_{24\text{ h}}$  on the first day of treatment and at steady state was used to assess the dosing regimen, aiming for  $>90\%$  PTA. When PTA for the target of  $AUC_{24\text{ h}} >400\text{ mg}\cdot\text{h/L}$  was not achieved, higher doses up to 1600 mg daily were explored.<sup>35</sup>

## Results

### Data

In total, 421 fluconazole concentrations from 25 Caucasian subjects (48% female), of which 17 were obese and 8 non-obese, with a TBW ranging from 61.0 to 174 kg, were included for pharmacokinetic analysis. One obese subject discontinued the study because of fluconazole extravasation during infusion (swelling disappeared within 24 h and no other abnormality was noted), whose concentrations measured upon oral dosing were included in the analysis. Subject details are presented in Table 1. All non-obese subjects and one obese subject had concentrations

obtained until 48 h after the first dose; the remaining obese patients had observations up to 24 h. Figure 1 shows the obtained fluconazole concentration–time profiles.

### Population pharmacokinetic model

Three absorption transit compartments connected by a one-compartment disposition model with first-order elimination and a combined proportional and additive residual error model best described the data.<sup>22</sup> Covariance values between parameters were all lower than 0.8. IIV was included on  $F$ ,  $V_d$  and the first-order rate constant between absorption transit compartments. No statistically significant influence of obesity or body size descriptors were found on  $F$ , therefore in the final model, the same  $F$  of 87.5% was estimated for obese and non-obese groups. IIV was relatively high, described as a 95% distribution interval of 43.9%–98.4%. Parameter estimates of the final model are presented in Table 2.

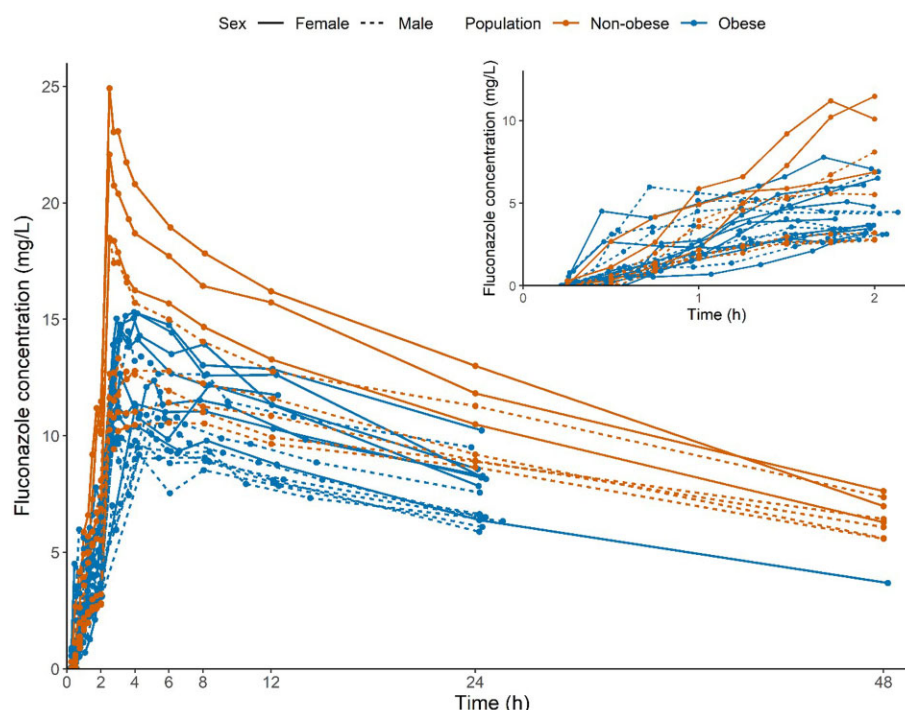
TBW in a power function in combination with sex presented a similar potential to describe IIV on  $V_d$  as LBW in a power function, yielding an OFV reduction of 45.1 versus 44.8 and a reduction of IIV on  $V_d$  from 25.2% to 6.80% versus 6.70%, respectively, with no discernible difference in GOF plots. The covariate function based on TBW and sex was included in the final model, as these

**Table 1.** Patient and data characteristics of the obese and non-obese subjects included in the pharmacokinetic analysis

| Characteristic                          | Obese            | Non-obese        |
|-----------------------------------------|------------------|------------------|
| No. of subjects                         | 17               | 8                |
| Sex, n (%)                              |                  |                  |
| Male                                    | 8 (47)           | 5 (63)           |
| Female                                  | 9 (53)           | 3 (37)           |
| Demographics, median (range)            |                  |                  |
| Age, years                              | 44 (25–62)       | 35 (23–60)       |
| TBW, kg                                 | 148 (106–174)    | 77.2 (61.0–93.5) |
| BMI, kg/m <sup>2</sup>                  | 44.1 (37.6–57.2) | 23.7 (19.0–26.9) |
| BSA, m <sup>2</sup>                     | 2.54 (2.11–2.82) | 1.95 (1.69–2.19) |
| LBW, kg                                 | 75.0 (53.8–88.7) | 60.0 (40.2–69.4) |
| Ideal bodyweight, kg                    | 72.3 (55.1–83.1) | 73.4 (58.7–81.3) |
| Adjusted bodyweight, kg                 | 101 (76.6–114)   | 75.8 (60.2–85.9) |
| Renal function measures, median (range) |                  |                  |
| Serum creatinine, $\mu\text{mol/L}$     | 75.0 (54.0–89.0) | 78.5 (70.0–91.0) |
| GFR, mL/min <sup>a</sup>                | 144 (109–187)    | 141 (86.7–164)   |
| Estimated GFR                           |                  |                  |
| CKD-EPI, mL/min/1.73 m <sup>2</sup>     | 143 (108–175)    | 106 (83.1–145)   |
| De-indexed CKD-EPI, mL/min <sup>b</sup> | 94.9 (79.7–120)  | 94.5 (79.0–120)  |
| MDRD, mL/min/1.73 m <sup>2</sup>        | 133 (102–178)    | 101 (77.7–147)   |
| De-indexed MDRD, mL/min <sup>b</sup>    | 94.0 (74.0–116)  | 90.0 (76.0–122)  |
| Cockcroft–Gault with TBW, mL/min        | 222 (143–290)    | 108 (74.2–161)   |
| Cockcroft–Gault with LBW, mL/min        | 105 (72.6–143)   | 76.1 (51.3–126)  |
| Sampling profile                        |                  |                  |
| No. of samples                          | 277              | 144              |
| No. of samples/subject, median (range)  | 16.3 (8–18)      | 18.0 (18–18)     |

<sup>a</sup>GFR was calculated based on 24 h urine.<sup>13</sup>

<sup>b</sup>De-indexed for BSA by multiplying the conventional values (in mL/min/1.73 m<sup>2</sup>) by BSA/1.73.



**Figure 1.** Individual concentration–time profiles of fluconazole for non-obese healthy subjects ( $n=8$ , orange) and obese but otherwise healthy subjects ( $n=17$ , blue). The upper right insert zooms in on the concentration–time profile in the first 2 h after oral dosing before IV administration. This figure appears in colour in the online version of *JAC* and in black and white in the printed version of *JAC*.

are more readily available in clinical practice. Incorporating TBW on CL using a power function further improved the GOF plots and dropped the OFV by 17.7 points ( $P<0.001$ ). Figure 2 illustrates the influence of TBW and sex on  $V_d$ , and TBW on CL from the final model. Introducing kidney function indices or other demographics did not further improve the model.

GOF plots indicate good descriptive performance of the final model and are presented in Figure S1. The jackknife results show that exclusion of none of the individuals caused a  $>10\%$  change in pharmacokinetic parameter estimates, indicating an absence of influential individuals. The NPDE results, shown in Figure S2, indicate an accurate predictive performance of the final model regarding both the structural and stochastic model for obese and non-obese subjects.

### Dose evaluations

Simulation results of the recommended fluconazole IV dosage regimen for invasive candidiasis in Figure 3 indicate that heavier subjects have lower steady-state exposure compared with lighter subjects. Moreover, male subjects have lower exposure early after treatment initiation compared with female subjects of the same weight and it takes longer for male and heavier subjects to reach steady state in comparison with female and lighter subjects. Figure 4 presents the distribution of fluconazole  $AUC_{24h}$  versus TBW on Day 1 and Day 7. With this regimen, all female subjects and 90% of male subjects lighter than 140 kg achieved the target of  $AUC_{24h} > 200$  mg·h/L on Day 1, and all individuals achieved this target at steady state. However, only subjects

lighter than 80 kg obtained the target of  $AUC_{24h} > 400$  mg·h/L for the treatment of *Candida* meningitis or encephalitis at steady state.

To ensure that all subjects receiving fluconazole IV achieve the target of  $AUC_{24h} > 200$  mg·h/L on the first day of treatment, male subjects heavier than 140 kg need a higher loading dose of 600 mg twice daily, compared with 800 mg once daily (Figure 5). To achieve an  $AUC_{24h} > 400$  mg·h/L at steady state, the fixed IV maintenance dose has to be increased from 400 to 600 mg per day for all patients (Figure S3). To achieve this high target on the first day, doses above 1600 mg, which are deemed potentially unsafe, are needed for most patients, particularly for male subjects (Figure S3).

Figures 6 and 7 present simulation results for a commonly used oral dosing regimen prescribed for treating oropharyngeal or oesophageal candidiasis. Due to the high IIV on F, exposure is highly variable for all weights. With the lower dose and variable F, no obese individual achieved the target of  $AUC_{24h} > 200$  mg·h/L. Other frequently used fluconazole oral dosing regimens were evaluated and the results can be found in Figures S4–S7.

### Discussion

This study shows that obesity alters fluconazole pharmacokinetics. The typical F of fluconazole capsules (87.5%) is within the reported range of 78%–162%,<sup>36</sup> and no statistically significant difference was identified on the F or absorption rate constant between the obese and non-obese, indicating that obesity has a very limited influence on the rate and extent of absorption of

**Table 2.** Pharmacokinetic parameter estimates for the final model

| Parameter                                                                                                                                           | Estimated values (RSE %) [95% CI] |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| $F = \frac{e^{\frac{\ln\left(\frac{\theta_F}{1-\theta_F}\right)}}}{1+e^{\frac{\ln\left(\frac{\theta_F}{1-\theta_F}\right)}}$ $\theta_F \text{ (%)}$ | 87.5 (7.70) [77.9–96.2]           |
| $k_{tr} \text{ (h}^{-1}\text{)}$                                                                                                                    | 2.69 (10.5) [2.30–3.22]           |
| $CL = CL_{70\text{ kg}} \times \left(\frac{TBW}{70}\right)^{\theta_{TBW,CL}}$ $CL_{70\text{ kg}} \text{ (L/h)}$                                     | 0.908 (3.00) [0.868–0.945]        |
| $\theta_{TBW,CL}$                                                                                                                                   | 0.390 (20.4) [0.214–0.532]        |
| $V_{d,female} = V_{d,70\text{ kg}} \times \left(\frac{TBW}{70}\right)^{\theta_{TBW,V_d}}$                                                           |                                   |
| $V_{d,male} = V_{d,70\text{ kg}} \times \left(\frac{TBW}{70}\right)^{\theta_{TBW,V_d}} \times (1 + \theta_{sex})$ $V_{d,70\text{ kg}} \text{ (L)}$  | 38.5 (2.50) [36.6–40.6]           |
| $\theta_{TBW,V_d}$                                                                                                                                  | 0.567 (10.6) [0.474–0.679]        |
| $\theta_{sex}$                                                                                                                                      | 0.269 (21.0) [0.177–0.364]        |
| IIV in coefficient of variation (%) (RSE%)                                                                                                          |                                   |
| $k_{tr}$                                                                                                                                            | 45.9 (14.0) [32.9–57.9]           |
| $V_d$                                                                                                                                               | 7.26 (23.8) [4.42–10.2]           |
| $\theta_F^a$                                                                                                                                        | 158 (25.7) [79.9–320]             |
| Residual error in variance (RSE%)                                                                                                                   |                                   |
| $\sigma_{prop}$                                                                                                                                     | 0.0033 (33.0) [0.00163–0.00522]   |
| $\sigma_{addi} \text{ (mg/L)}$                                                                                                                      | 0.266 (30.5) [0.180–0.397]        |

RSE, relative standard error; CI, CI obtained in the sampling importance resampling procedure;  $\theta_F$ , population mean value of  $F$ ;  $k_{tr}$ , first-order rate constant between absorption transit compartments;  $CL_{70\text{ kg}}$ , the population mean value of  $CL$  for a subject with a weight of 70 kg;  $\theta_{TBW,CL}$ , exponent in the exponential covariate relationships between  $TBW$  and  $CL$ ;  $V_{d,female}$ ,  $V_d$  for female subjects;  $V_{d,male}$ ,  $V_d$  for male subjects;  $V_{d,70\text{ kg}}$  the population mean value of the  $V_d$  for a subject with the weight of 70 kg;  $\theta_{TBW,V_d}$ , exponent in the exponential covariate relationships between  $TBW$  and  $V_d$ ;  $\theta_{sex}$ , proportional increase in  $V_d$  for male compared with female subjects;  $\sigma_{prop}$ , proportional residual error;  $\sigma_{addi}$ , additive residual error.

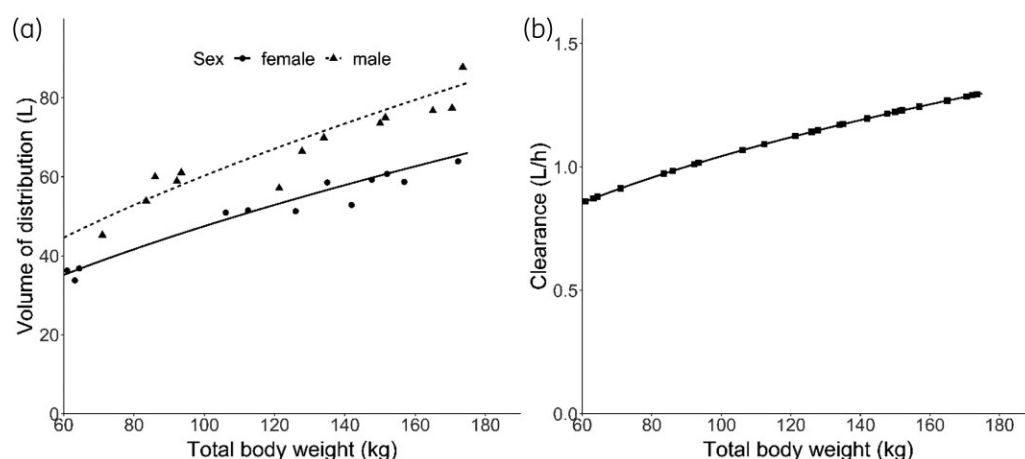
<sup>a</sup>Due to logit transformation, IIV of  $F$  could be described as 95% distribution interval with the 2.5th and 97.5th percentiles calculated by

$$\left( \frac{e^{\frac{\ln\left(\frac{\theta_F}{1-\theta_F} - 1.96 \times \sqrt{\omega_F^2}\right)}}}{1+e^{\frac{\ln\left(\frac{\theta_F}{1-\theta_F} - 1.96 \times \sqrt{\omega_F^2}\right)}}} \right) \text{ and } \left( \frac{e^{\frac{\ln\left(\frac{\theta_F}{1-\theta_F} + 1.96 \times \sqrt{\omega_F^2}\right)}}}{1+e^{\frac{\ln\left(\frac{\theta_F}{1-\theta_F} + 1.96 \times \sqrt{\omega_F^2}\right)}}} \right) \text{ i.e. 43.9\% and 98.4\%}.$$

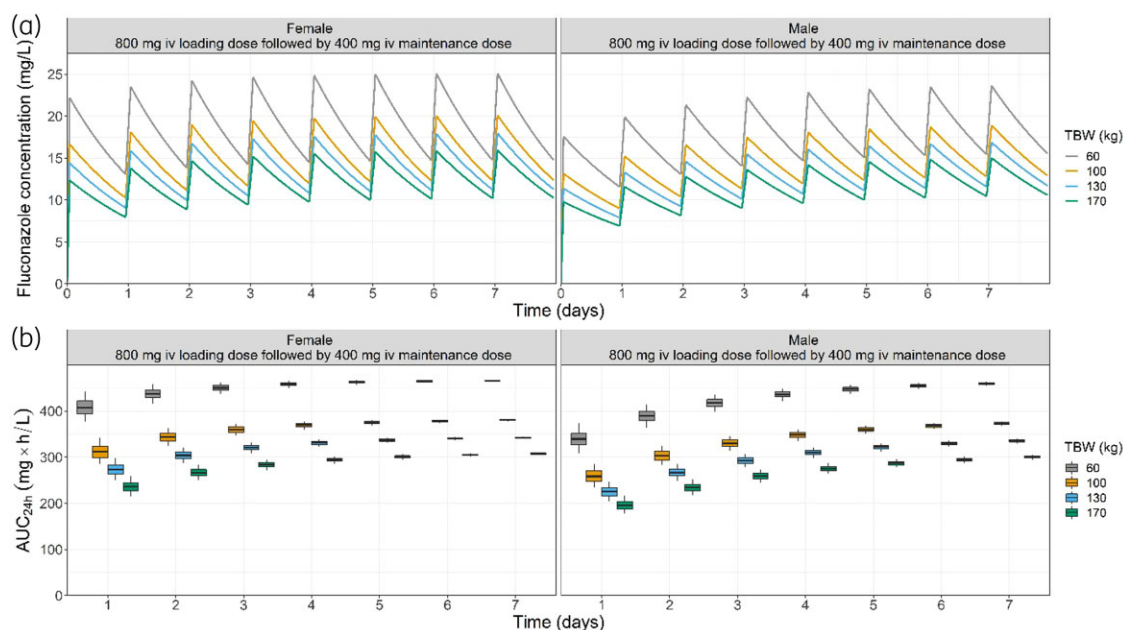
fluconazole capsules. Despite a high average  $F$  for fluconazole capsules, caution should be taken when switching from IV to the oral capsule due to the high IIV in  $F$ . In obese adults up to 174 kg,  $TBW$  is significantly correlated with  $V_d$  and  $CL$ , which can be described with power functions (Table 2). Additionally, the  $V_d$  in male subjects is on average 26.9% larger than in female subjects of the same weight.

Only a few studies investigated fluconazole in the obese population. Alobaid *et al.*<sup>37</sup> investigated the pharmacokinetics of fluconazole in critically ill obese patients. Although no statistically significant covariate relationship was found from this study, the measured  $CL_{CR}$  was included as a covariate on  $CL$ , merely due to the improvement in the diagnostic plots and, similarly,  $BMI$  was used as a descriptor for  $V_d$  of the central compartment based on biological plausibility and improvement from the diagnostic plots. The small sample size in this study and the pathophysiological complexity of critically ill patients might have obscured the impact of obesity. An important strength of our study is the prospective study design with semi-simultaneous oral and IV dosing, which allows for an accurate estimation of  $F$  by reducing the influence of inter-occasion variability, the intensive sampling and the wide range of  $TBW$ . By selecting relatively healthy

individuals, the potentially confounding influence of pathological factors such as renal dysfunction is circumvented; however, this comes with the limitation that extrapolations to patients, particularly patients with renal dysfunction or other relevant pathological factors, cannot be made directly. Although the bariatric surgery during this pharmacokinetic study might interfere with the pharmacokinetics, we anticipate that this influence may be negligible as the duration of this surgery is short (<1 h) with minor blood loss (<50 mL). Although the population  $CL$  in the healthy obese patients from our study is very similar to what has been reported in critically ill obese patients (0.908 versus 0.950 L/h), a high IIV of 50.5% on  $CL$  was found in those patients while no IIV on  $CL$  could be identified by our model,<sup>37</sup> which suggests it may be more challenging to dose critically ill obese patients with whom multiple comorbidities are commonly associated. Pharmacokinetics studies conducted in various patient populations identified that kidney function and disease severity are associated with fluconazole  $CL$ .<sup>37–40</sup> We have not found kidney function estimates to be statistically significant predictors of IIV, which is likely attributable to the absence of individuals with impaired renal function. Additionally, in our model, the fluconazole  $CL$  of a 70 kg healthy individual is 0.908 L/h,



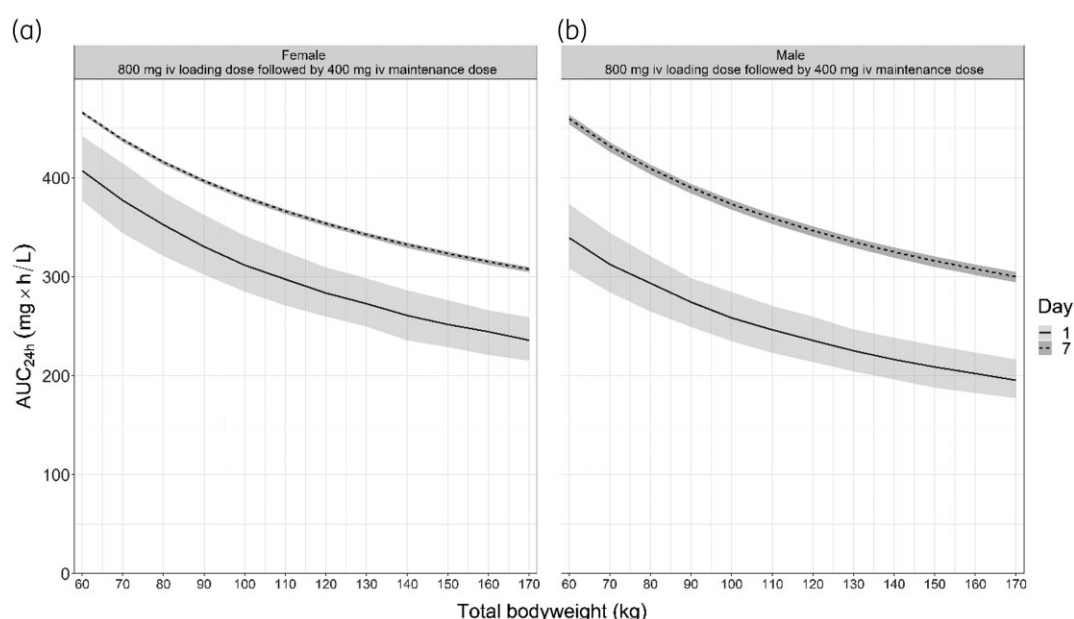
**Figure 2.** Individual empirical Bayes estimates (filled circles, filled triangles and filled squares) for the  $V_d$  (a) and CL (b) versus TBW from the final model. Lines represent the model-predicted relationships between the  $V_d$  and CL versus TBW.



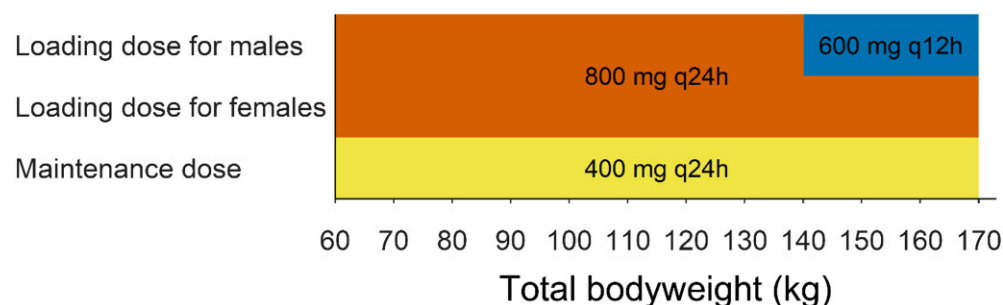
**Figure 3.** Median fluconazole concentration-time profiles (a) and distribution of the  $AUC_{24h}$  (b) based on 1000 simulations of female and male subjects of various TBWs receiving a recommended IV loading dose of 800 mg once daily followed by a maintenance dose of 400 mg once daily. The boxes represent the 25th, 50th (median) and 75th percentiles, and whiskers represent the 5th and 95th percentiles (i.e. 90% distribution interval). This figure appears in colour in the online version of JAC and in black and white in the printed version of JAC.

corresponding to 15.1 mL/min, which is much lower than the average GFR in our population. This is in line with previous reports on extensive passive tubular reabsorption of fluconazole.<sup>41</sup> Although concentrations at 48 h were mostly missing from the obese group, no clear change in the elimination profile was noticed from 24 to 48 h based on the available concentrations at 48 h. Additionally, the estimation results of the final model remain similar when all concentrations at 48 h were excluded. Therefore, we do not expect that these missing observations at 48 h would alter our findings.

In addition to TBW, we found sex to be correlated with  $V_d$ . Interestingly, as sex is incorporated in the calculation of LBW, a similar descriptive potential of IIV in  $V_d$  could be obtained with LBW in comparison with the combination of TBW and sex. The contribution of height in the calculation of LBW appears to be negligible in our analysis, possibly because the range in height covers a difference of less than 30 cm. We decided to include the combination of TBW and sex to facilitate the clinical implementation of model-derived dosing recommendations by avoiding complex calculations of LBW. Higher body fat composition,



**Figure 4.** Distribution of  $AUC_{24h}$  values versus TBW for fluconazole on Day 1 (solid line) and Day 7 (dashed line) based on 1000 simulations in female (a) and male (b) subjects receiving an IV loading dose of 800 mg once daily followed by a maintenance dose of 400 mg once daily. The shaded areas represent the 90% prediction interval.



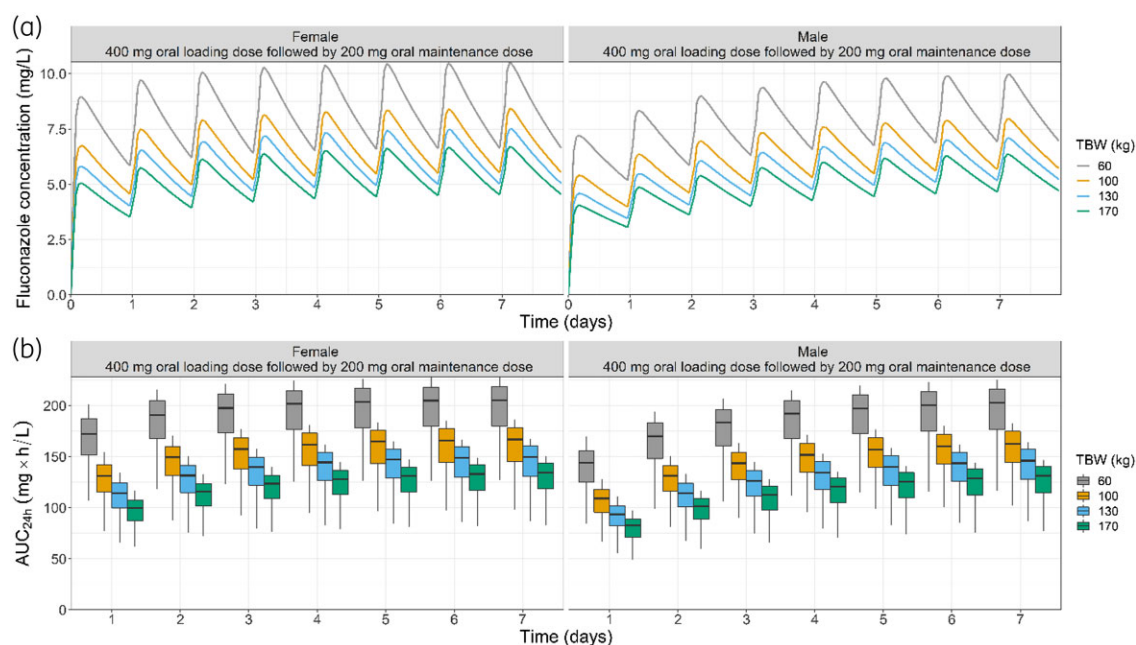
**Figure 5.** Model-derived IV loading and maintenance dose recommendations for fluconazole for achieving a target  $AUC_{24h} > 200$  mg-h/L for the first day of treatment in female and male subjects of various TBWs. This figure appears in colour in the online version of JAC and in black and white in the printed version of JAC.

namely a lower body water composition in female versus male subjects with the same TBW, could potentially explain the smaller  $V_d$  in female subjects.<sup>42</sup> Due to the increased  $V_d$ , obese male subjects with TBW  $\geq 140$  kg need an increased loading dose (Figure 5) to achieve target exposure on the first day of treatment.

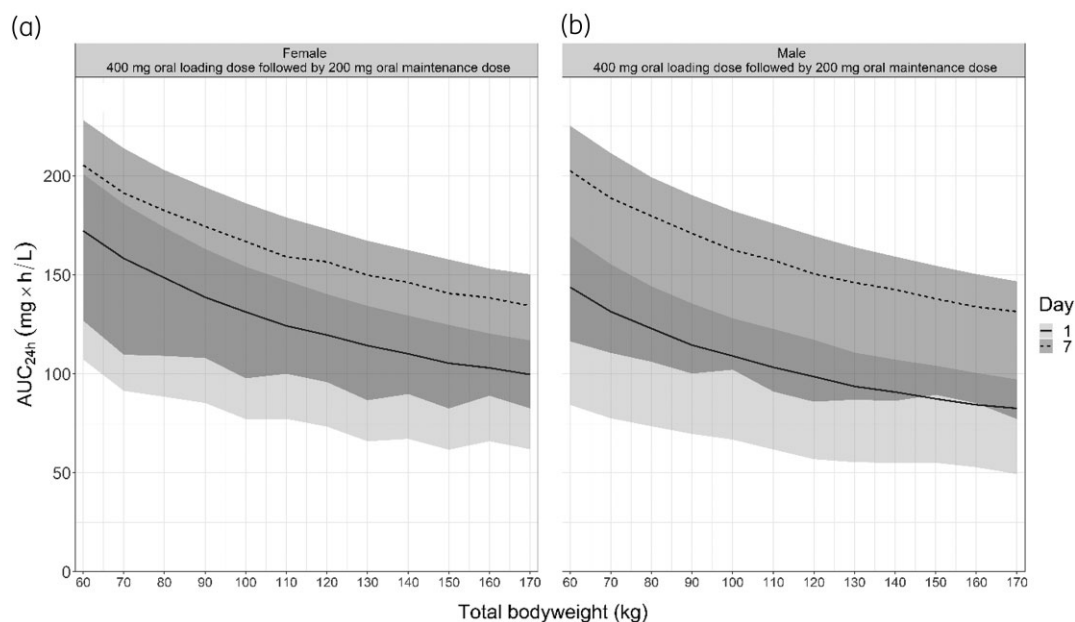
With the dosing recommendations for obese patients as derived in our study (Figure 5), the target of  $AUC_{24h}/MIC > 100$  for a pathogen with  $MIC \leq 2$  mg/L can be achieved. This recommendation is anticipated to be safe as a 1200 mg daily dose for 2 weeks has shown good tolerance and no liver function disturbance in 30 HIV patients.<sup>43</sup> A recent pharmacokinetic study in critically ill obese patients suggested a TBW-based loading dose of 12 mg/kg and a maintenance dose of 6 mg/kg for pathogens with an MIC of 2 mg/L. With this dosing strategy, a loading dose of  $>1600$  mg is required in patients with TBW  $> 130$  kg, which is partly unnecessary according to our simulation

(Figure 5) and potentially unsafe.<sup>35</sup> For the empirical treatment of *Candida* meningitis or encephalitis, a higher exposure might be desirable to compensate for the 20%–50% reduced fluconazole penetration into the CSF.<sup>32,33</sup> Therefore we also assessed PTA for an  $AUC_{24h}$  target of 400 mg-h/L, and the simulation results indicate that an increased maintenance dose of 600 versus 400 mg once daily is required, and loading doses exceeding 1600 mg for female subjects  $\geq 140$  kg and male subjects  $\geq 90$  kg to meet this target on Day 1 (Figure S3).<sup>35</sup> Clinicians should balance potential fluconazole-related toxicity with the decreased PTA when treating obese patients with *Candida* infections in the CNS.

We investigated the exposure levels for three commonly used fluconazole oral regimens including a loading (first day)/maintenance dose of 400/200 mg, 200/100 mg and 150 mg every third day for a total of three doses (Day 1, 4 and 7) in the first week, and 150 mg weekly (Figures 6–7 and Figures S4–S7), which



**Figure 6.** Median fluconazole concentration–time profiles (a) and distribution of the AUC<sub>24 h</sub> (b) based on 1000 simulations of female and male subjects of various TBW receiving a recommended oral loading dose of 400 mg once daily followed by a maintenance dose of 200 mg once daily. The boxes represent the 25th, 50th (median) and 75th percentiles, and whiskers represent the 5th and 95th percentiles (i.e. 90% distribution interval). This figure appears in colour in the online version of JAC and in black and white in the printed version of JAC.



**Figure 7.** Distribution of AUC<sub>24 h</sub> values versus TBW for fluconazole on Day 1 (solid line) and Day 7 (dashed line) based on 1000 simulations in female (a) and male (b) subjects receiving an oral loading dose of 400 mg once daily followed by a maintenance dose of 200 mg once daily. The shaded areas represent the 90% prediction interval.

are primary treatments for superficial and mucosal *Candida* infections. Hardly any individual with TBW between 80 and 170 kg reached the AUC<sub>24 h</sub> > 200 mg·h/L target with these

doses, yet favourable clinical responses have been reported, suggesting that a lower target exposure may be effective.<sup>44,45</sup> This could potentially be explained by the sufficient penetration of

fluconazole.<sup>46</sup> Alternatively, the susceptibility of *Candida* spp. to fluconazole could be increased, or the immune response is more active with these infections. Symptomatic relapse of vulvo-vaginal candidiasis was reported in approximately 40% of women,<sup>31</sup> while recurrent oropharyngeal and oesophageal candidiasis have also become an increasingly prevalent clinical issue.<sup>44</sup> Potentially these refractory superficial infections might result from the highly variable *F* we observed in obese and non-obese individuals, which means that a certain proportion of patients can be underexposed because they have a low *F* that could even exacerbate the infection by selecting more resistant strains of *Candida* spp.

In conclusion, our results show that in otherwise healthy obese adults, both fluconazole CL and *V<sub>d</sub>* increase with increasing TBW, with sex being an additional covariate for the *V<sub>d</sub>*, resulting in a larger *V<sub>d</sub>* in male compared with female subjects of the same weight. As a result, male subjects with high TBW may need increased loading doses as the time to steady state is longer. Model-based evaluations of commonly used oral dosing regimens illustrate high variability in exposure due to the high IIV in *F*, which could put large proportions of obese individuals at higher risk of underexposure.

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## Transparency declarations

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## Supplementary data

Tables S1 and S2, Figures S1 to S7 and NONMEM control stream are available as [Supplementary data](#) at JAC Online.

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